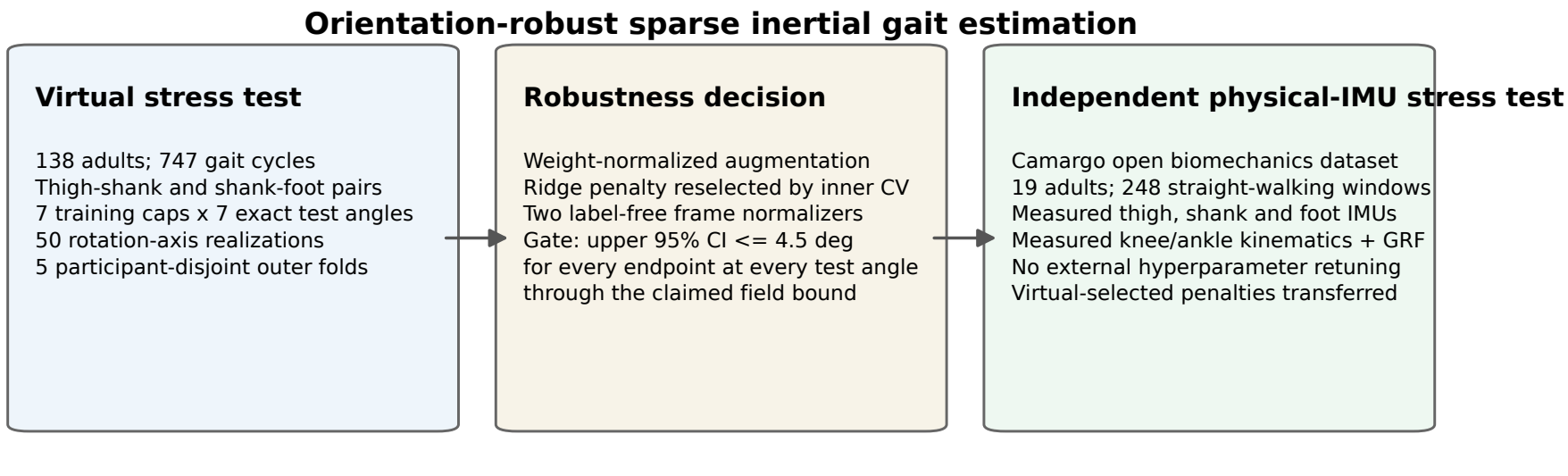
Orientation-robust sparse inertial gait estimation
Virtual stress test
138 adults; 747 gait cycles
Thigh-shank and shank-foot pairs
7 training caps x 7 exact test angles
50 rotation-axis realizations
5 participant-disjoint outer folds
Robustness decision
Weight-normalized augmentation
Ridge penalty reselected by inner CV
Two label-free frame normalizers
Gate: upper 95% CI <= 4.5 deg
for every endpoint at every test angle
through the claimed field bound
Independent physical-IMU stress test
Camargo open biomechanics dataset
19 adults; 248 straight-walking windows
Measured thigh, shank and foot IMUs
Measured knee/ankle kinematics + GRF
No external hyperparameter retuning
Virtual-selected penalties transferred